musicar: MIDI generation transformers

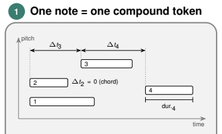
Three decoder-only transformers over compound note tokens (one position = one note, 20 ms grid), pretrained without captions. Each diagram has its own key.

musicar-L (464M)

Pretrained multi-instrument MIDI model without text captions: 464.24M parameters (trunk 445.98M, cascade heads 16.28M, embeddings 1.99M), trained on 8.3B notes.

How to read: numbers give the reading order; teal = the model's structure; grey boxes = linear layers and embedding tables; dashed frame = a module drawn open in place; dotted frame or box = a zoom into the dotted part.

1 One note = one compound token

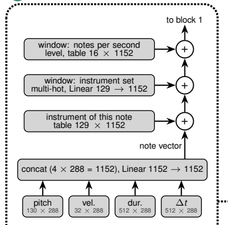


Notes 1–4 become positions 1–4: one position per note, in onset order, each with all five values:

field	values	meaning
Δ*t*	512	time since the previous note started (20 ms steps)
instr.	129	General MIDI program 0–127 or drums
pitch	130	MIDI pitch 0–127 + start / end of a piece (BOS / EOS)
dur.	512	note length (20 ms steps)
vel.	32	velocity: how hard the note is played

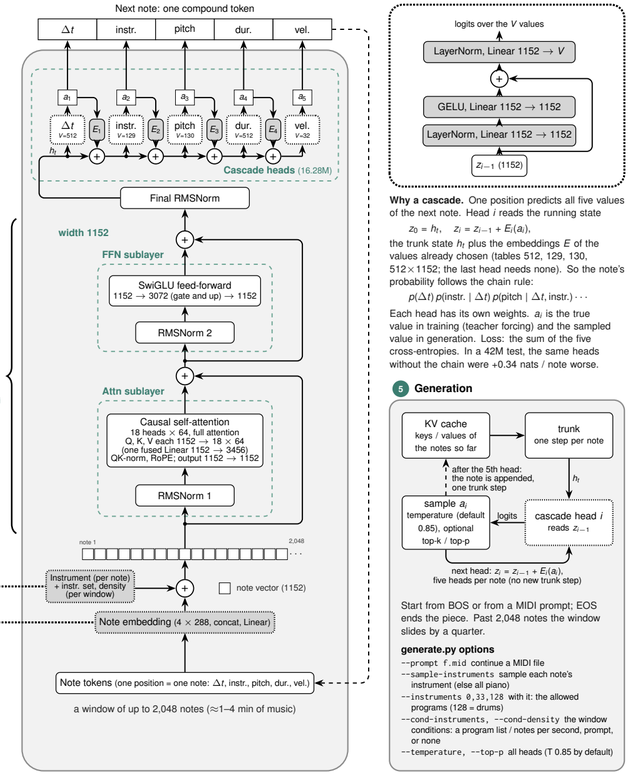
Time is absolute: no bars, beats or tempo tokens; Δ*t* and durations stop at 10.22 s. The notes of all tracks are merged into one sequence; for drums the pitch is the drum sound.

2 Input embedding (1.99M)



The instr. head predicts the next note's instrument (128 General MIDI programs + drums). The two **window conditions** are added at every position, measured on the window's own notes (density: 15 levels + "none"). In training both are replaced by "none" in 15% of the windows, so "none" (unconditional) stays usable; in generation they come from the user or the prompt. The instrument table and both conditions start at zero.

3 Music model



a window of up to 2,048 notes (≈1–4 min of music)

4 One cascade head (one per dotted box)

Why a cascade. One position predicts all five values of the next note. Head *i* reads the running state

$$z_0 = h_t, \quad z_i = z_{i-1} + E_i(a_i),$$

the trunk state h_t plus the embeddings E of the values already chosen (tables 512, 129, 130, 512×1152; the last head needs none). So the note's probability follows the chain rule:

$$p(\Delta t)\, p(\text{instr.} \mid \Delta t)\, p(\text{pitch} \mid \Delta t, \text{instr.}) \cdots$$

Each head has its own weights. a_i is the true value in training (teacher forcing) and the sampled value in generation. Loss: the sum of the five cross-entropies. In a 42M test, the same heads without the chain were +0.34 nats / note worse.

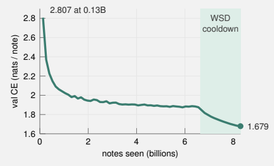
5 Generation

Start from BOS or from a MIDI prompt; EOS ends the piece. Past 2,048 notes the window slides by a quarter.

generate.py options

- `--prompt f.mid` continue a MIDI file
- `--sample-instruments` sample each note's instrument (else all piano)
- `--instruments 0,33,128` with it: the allowed programs (128 = drums)
- `--cond-instruments`, `--cond-density` the window conditions: a program list / notes per second, prompt, or none
- `--temperature`, `--top-p` all heads (T 0.85 by default)

6 Pretraining



Every 2,000 steps (65,536 notes per step): the sum of the Δ*t*, pitch, duration and velocity cross-entropies on GigaMIDI validation files whose notes do not occur in its training split. Constant LR, then the $1 - \sqrt{\cdot}$ cooldown in the last 20%. One run.

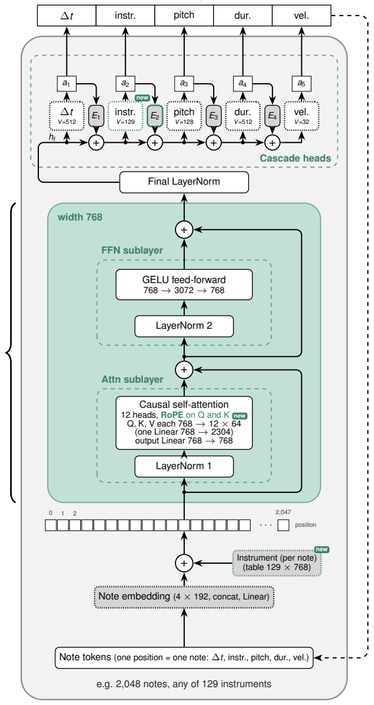
Setting	Base pretraining (done)
Data	GigaMIDI 0.4, Aria-MIDI 0.3, Discover MIDI 0.3 (sampling weights); no audio-derived songs
Notes seen	8.3B (≈1.8 epochs on average; Aria-MIDI ≈4.5)
Steps, batch	126,582 steps × 65,536 notes (32 windows of 2,048 notes)
Optimizer	Muon (Moonlight scaling) for the weight matrices of the blocks, AdamW for the rest; one LR 10^{-3}, weight decay 0.1
Schedule	WSD: 1,000 warmup steps, constant LR, then $1 - \sqrt{\cdot}$ decay to 0 over the last 20%
Loss	sum of the five cross-entropies, teacher forcing; no dropout, no label smoothing
Augmentation	random transposition by up to ±5 semitones (drums unchanged)
Conditions	instrument set and density replaced by "none" in 15% of the windows
Precision	bf16 with fp8 matmuls, torch.compile; the checkpoint holds bf16 weights
Hardware	one rented RTX 5090, ≈41 h
Result	clean GigaMIDI validation **1.679** nats / note = Δ*t* 0.170 + pitch 0.367 + dur. 0.644 + vel. 0.497 (the instrument head, 0.117, apart); 1.681 with both conditions "none"

musicar-S (108M)

107.93M parameters: trunk 99.11M, cascade heads 7.90M, embeddings 0.92M. 129 instruments, context 2,048 notes.

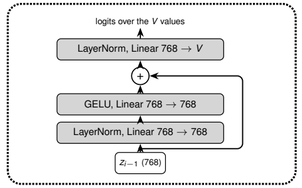
How to read: numbers give the reading order; teal = the repeated block; grey boxes = linear layers and embedding tables; dashed frame = a module drawn open in place; dotted frame or box = a zoom (white panel) into the dotted part; new and teal outlines = added when the piano model was grown (section 4).

1 Music model



e.g. 2,048 notes, any of 129 instruments

3 One cascade head



Why a cascade. Head *i* (a dotted box, own weights) reads the trunk state plus the embeddings E of the values already chosen,

$$z_0 = h_t, \quad z_i = z_{i-1} + E_i(a_i),$$

so the note's probability follows the chain rule:
$p(\Delta t)\, p(\text{instr.} \mid \Delta t) \cdots$. In training a_i is the true value (teacher forcing), in generation the sampled one. Loss: the sum of the five cross-entropies.

4 Grown from musicar-S-piano

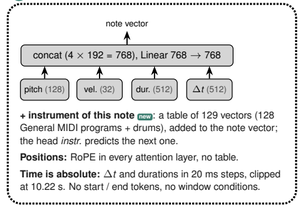
Copied (and trained on): the trunk, the note embedding, the four note heads and their condition embeddings.

New: the instrument embedding, the instrument head and its condition embedding E_2; both tables and the head's output layer start at zero: at step 0 they change nothing.

Changed: learned positions → RoPE, context 512 → 2,048 notes (not function-preserving).

Run: GigaMIDI 0.7 + Aria-MIDI 0.3, 197M notes (3,000 steps × 65,536), 83 min. GigaMIDI validation 2.232 nats / note; the same recipe from scratch: 2.597.

2 Input embedding (the dotted boxes)



+ instrument of this note new: a table of 129 vectors (128 General MIDI programs + drums), added to the note vector; the head *instr.* predicts the next one.

Positions: RoPE in every attention layer, no table.

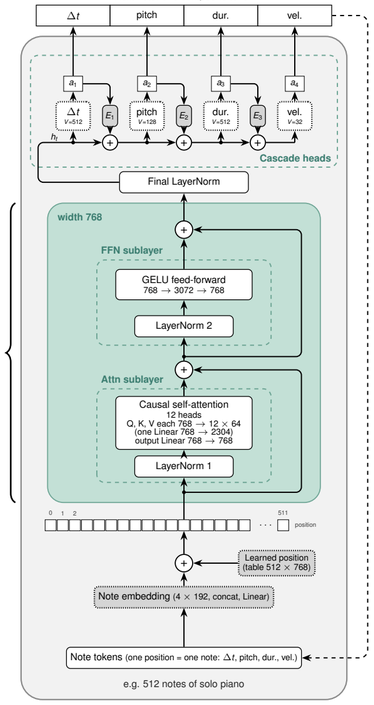
Time is absolute: Δ*t* and durations in 20 ms steps, clipped at 10.22 s. No start / end tokens, no window conditions.

musicar-S-piano (107M)

106.84M parameters: trunk 99.11M, cascade heads 6.52M, embeddings 1.21M. Solo piano, context 512 notes.

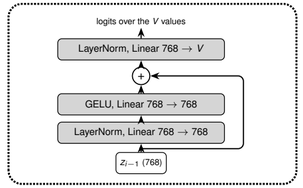
How to read: numbers give the reading order; teal = the repeated block; grey boxes = linear layers and embedding tables; dashed frame = a module drawn open in place; dotted frame or box = a zoom (white panel) into the dotted part.

1 Music model



e.g. 512 notes of solo piano

3 One cascade head



Why a cascade. Head *i* (a dotted box, own weights) reads the trunk state plus the embeddings E of the values already chosen,

$$z_0 = h_t, \quad z_i = z_{i-1} + E_i(a_i),$$

so the note's probability follows the chain rule:
$p(\Delta t)\, p(\text{pitch} \mid \Delta t) \cdots$. In training a_i is the true value (teacher forcing), in generation the sampled one. Loss: the sum of the four cross-entropies.

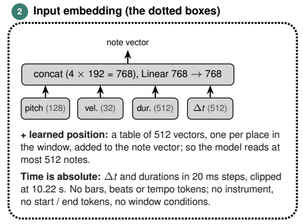
4 Training (done, 2026-09-30)

Data: MAESTRO + GiantMIDI-Piano (≈5% of the notes) and Aria-MIDI (≈95%), solo piano.

Run: 38,000 steps × 61,440 notes (120 windows of 512) = 2.33B notes, ≈4 passes; Muon + AdamW, LR 10^{-3}, WSD schedule; 12 h 27 min on a laptop RTX 5060 (8 GB).

Result: Aria-MIDI validation 5.706 nats / note, inside the 5.47–5.72 forecast of the scaling ladder; MAESTRO + GiantMIDI-Piano validation 6.987.

2 Input embedding (the dotted boxes)



+ learned position: a table of 512 vectors, one per place in the window, added to the note vector; so the model reads at most 512 notes.

Time is absolute: Δ*t* and durations in 20 ms steps, clipped at 10.22 s. No bars, beats or tempo tokens; no instrument, no start / end tokens, no window conditions.

Comparison

	musicar-L	musicar-S	musicar-S-piano
Parameters	464.24M: trunk 445.98M, cascade heads 16.28M, embeddings 1.99M	107.93M: trunk 99.11M, cascade heads 7.90M, embeddings 0.92M	106.84M: trunk 99.11M, cascade heads 6.52M, embeddings 1.21M
Trunk	28 blocks, width 1152, 18 attention heads	14 blocks, width 768, 12 attention heads	14 blocks, width 768, 12 attention heads
Context	2,048 notes	2,048 notes	512 notes
Instruments	129 (128 General MIDI programs + drums), BOS / EOS, window conditions (instrument set, density)	129 (128 General MIDI programs + drums); no BOS / EOS, no conditions	piano only; no BOS / EOS, no conditions
Positions	RoPE	RoPE (replaced the piano model's learned table)	learned table (512 × 768)
Norm / MLP	RMSNorm, SwiGLU, QK-norm	LayerNorm, GELU	LayerNorm, GELU
Data	GigaMIDI 0.4, Aria-MIDI 0.3, Discover MIDI 0.3	grown from musicar-S-piano, then GigaMIDI 0.7, Aria-MIDI 0.3	MAESTRO + GiantMIDI-Piano ≈5%, Aria-MIDI ≈95%
Notes seen	8.3B (126,582 steps, ≈41 h on one RTX 5090)	197M (3,000 steps, 83 min on a laptop RTX 5060), after the piano model's 2.33B	2.33B (38,000 steps, 12 h 27 min on a laptop RTX 5060)
Result	clean GigaMIDI validation **1.679**	GigaMIDI validation **2.232** (older split that shares files with training)	Aria-MIDI validation **5.706** (512-note windows)

Result: validation cross-entropy in nats per note (teacher-forced, summed over Δ*t*, pitch, duration and velocity; lower is better). The three validation sets differ, so the numbers are not comparable across columns.

Next (planned, not done)

